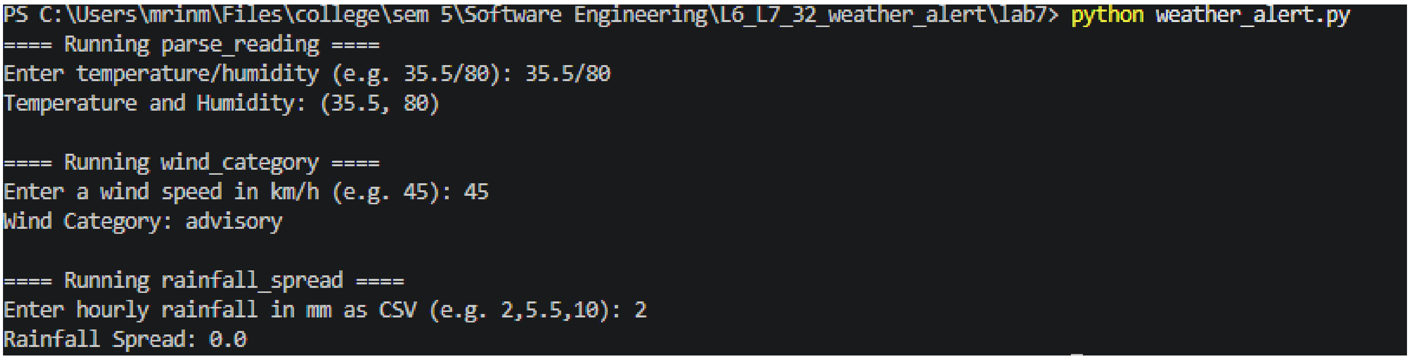
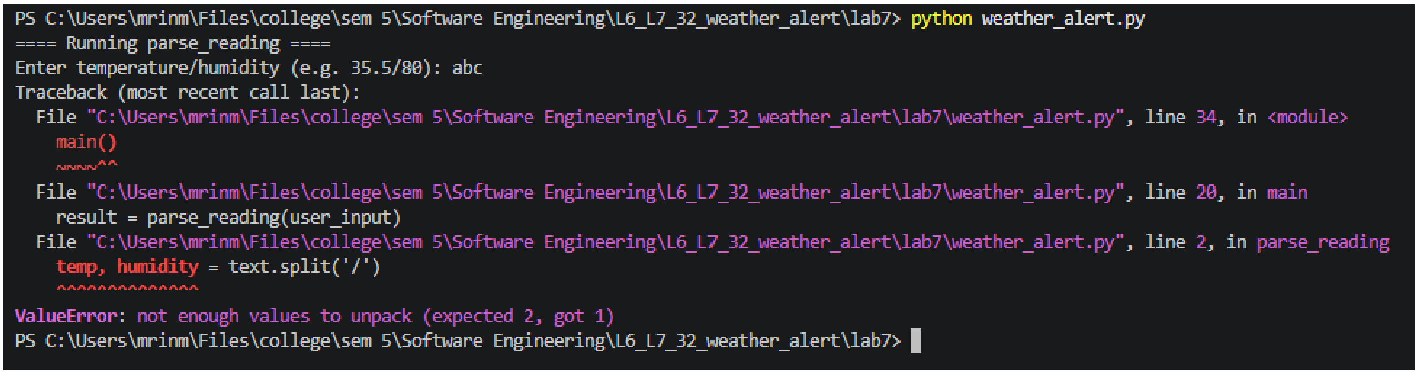
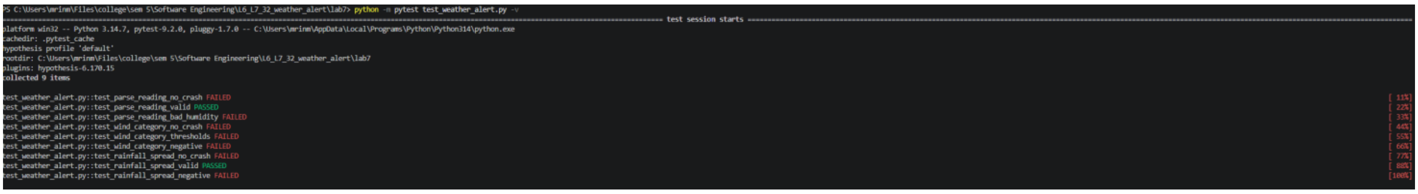
Original application:



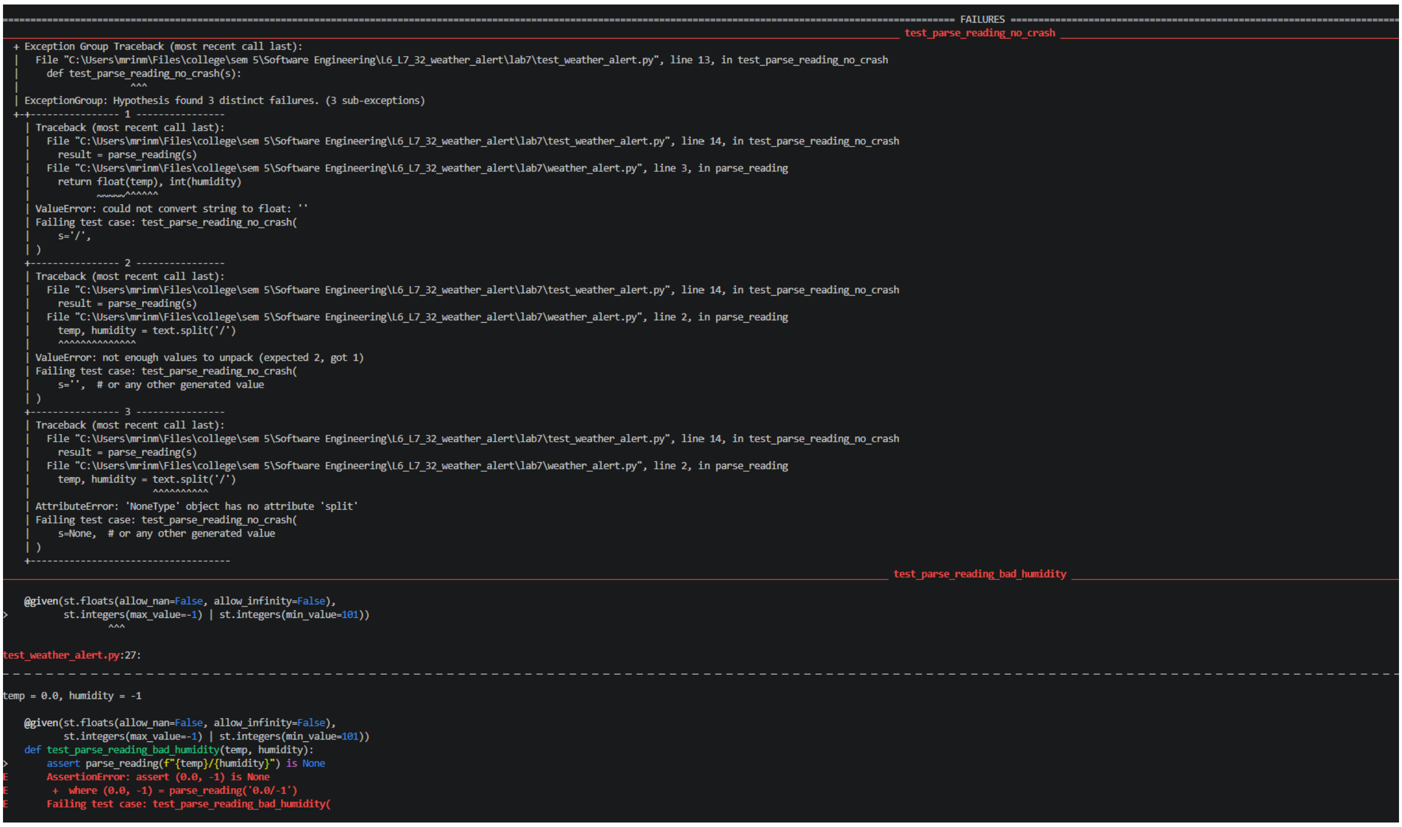
```
PS C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7> python weather_alert.py
==== Running parse_reading ====
Enter temperature/humidity (e.g. 35.5/80): 35.5/80
Temperature and Humidity: (35.5, 80)

==== Running wind_category ====
Enter a wind speed in km/h (e.g. 45): 45
Wind Category: advisory

==== Running rainfall_spread ====
Enter hourly rainfall in mm as CSV (e.g. 2,5.5,10): 2
Rainfall Spread: 0.0
```



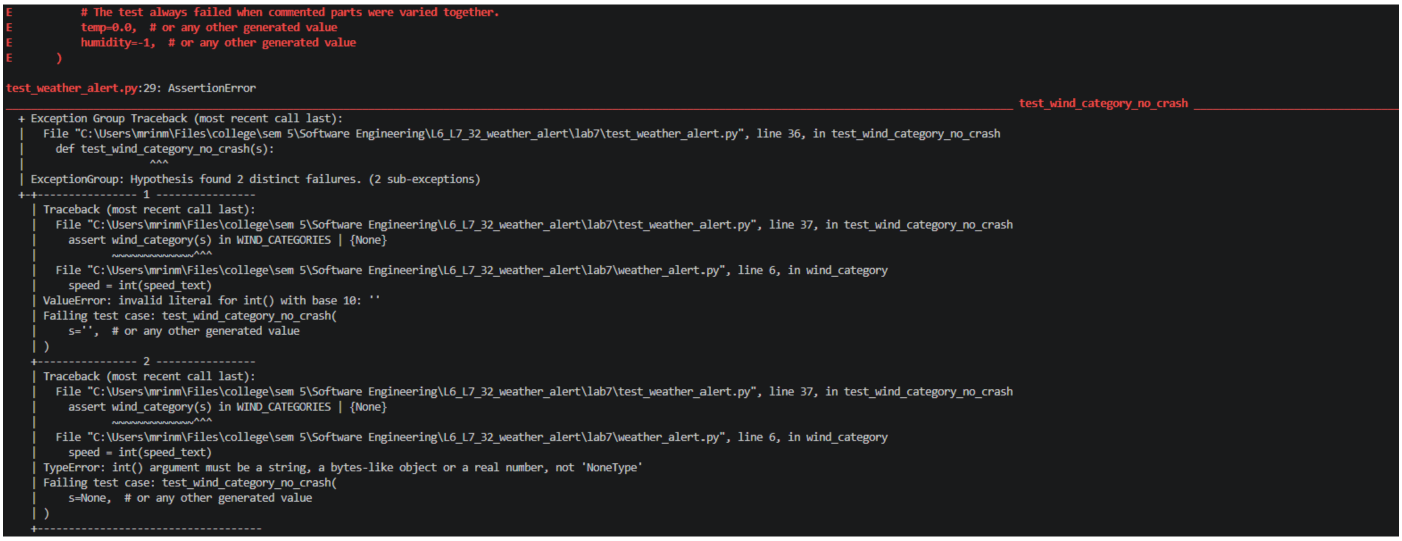
```
PS C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7> python weather_alert.py
==== Running parse_reading ====
Enter temperature/humidity (e.g. 35.5/80): abc
Traceback (most recent call last):
  File "C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7\weather_alert.py", line 34, in <module>
    main()
    ~~~~^^
  File "C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7\weather_alert.py", line 20, in main
    result = parse_reading(user_input)
  File "C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7\weather_alert.py", line 2, in parse_reading
    temp, humidity = text.split('/')
    ^^^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)
PS C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7>
```



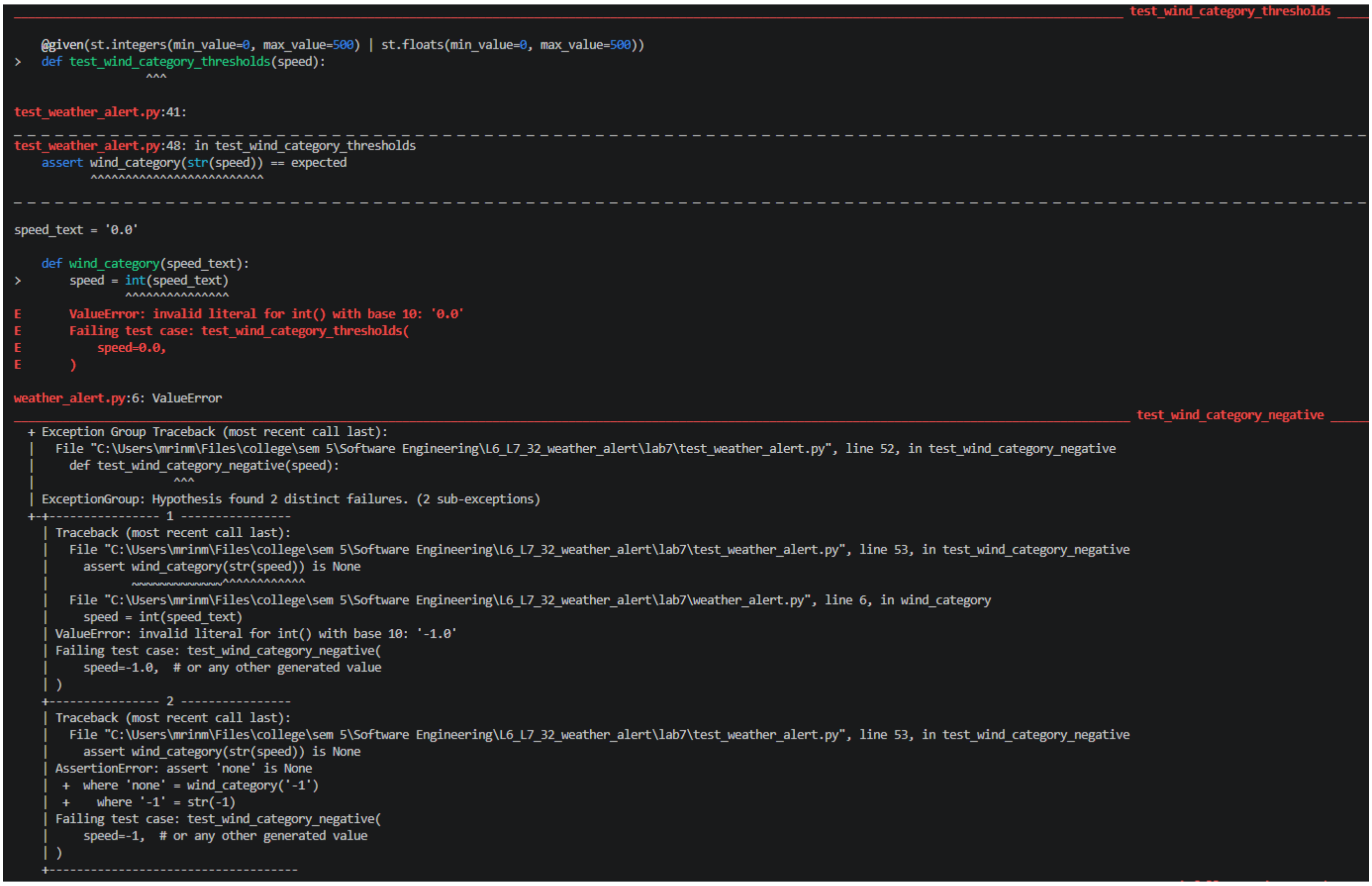
```
PS C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7> python -m pytest test_weather_alert.py -v
==================================================================================================== test session starts ====================================================================================================
platform win32 -- Python 3.14.7, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\mrinm\AppData\Local\Programs\Python\Python314\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7
plugins: hypothesis-6.170.15
collected 9 items

test_weather_alert.py::test_parse_reading_no_crash FAILED                                                                                                                                                              [ 11%]
test_weather_alert.py::test_parse_reading_valid PASSED                                                                                                                                                                 [ 22%]
test_weather_alert.py::test_parse_reading_bad_humidity FAILED                                                                                                                                                          [ 33%]
test_weather_alert.py::test_wind_category_no_crash FAILED                                                                                                                                                              [ 44%]
test_weather_alert.py::test_wind_category_thresholds FAILED                                                                                                                                                            [ 55%]
test_weather_alert.py::test_wind_category_negative FAILED                                                                                                                                                              [ 66%]
test_weather_alert.py::test_rainfall_spread_no_crash FAILED                                                                                                                                                            [ 77%]
test_weather_alert.py::test_rainfall_spread_valid PASSED                                                                                                                                                               [ 88%]
test_weather_alert.py::test_rainfall_spread_negative FAILED                                                                                                                                                            [100%]
```



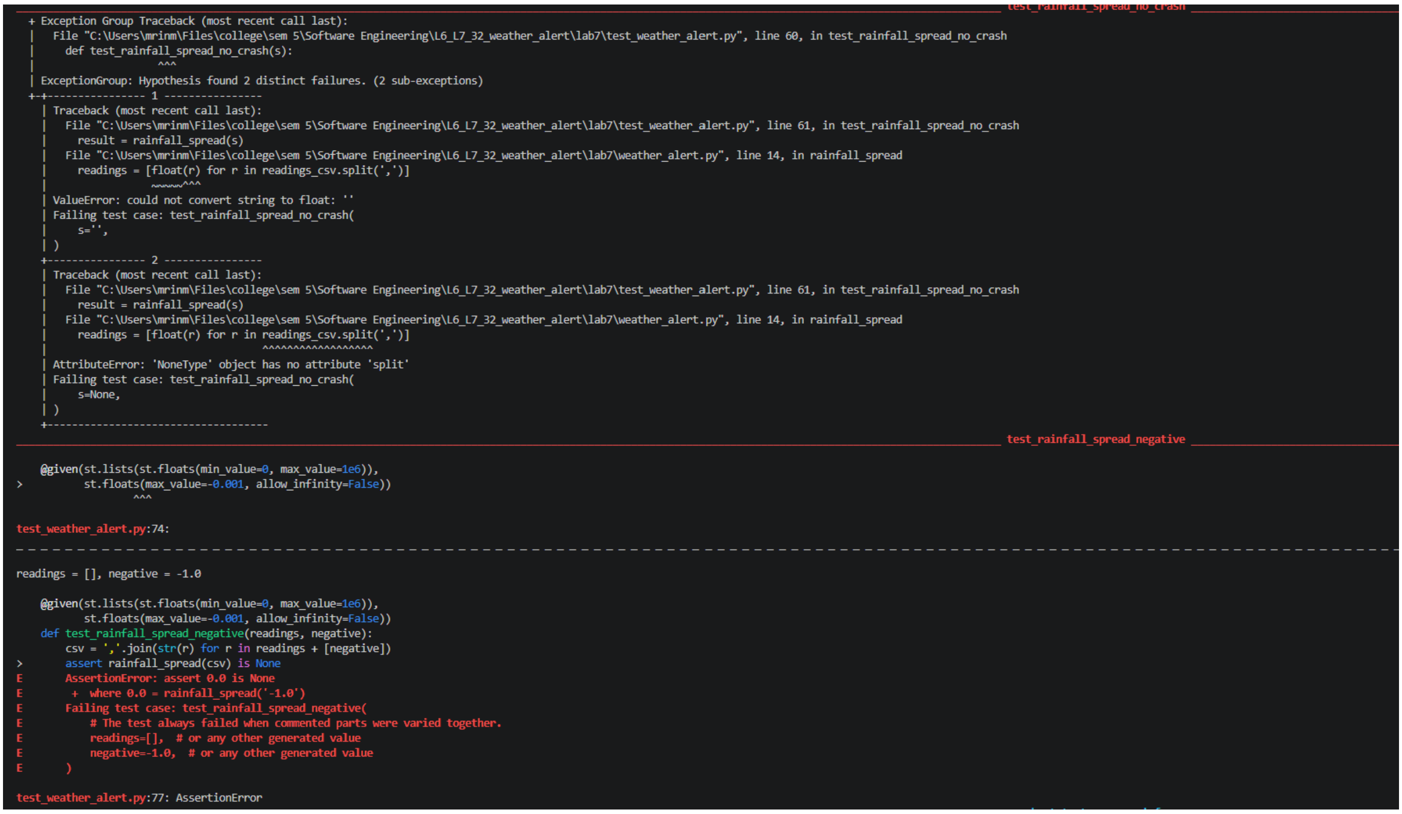
```
========================================================================================================= FAILURES =========================================================================================================
__________________________________________________________________________________________________ test_parse_reading_no_crash ___________________________________________________________________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7\test_weather_alert.py", line 13, in test_parse_reading_no_crash
  |     def test_parse_reading_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7\test_weather_alert.py", line 14, in test_parse_reading_no_crash
    |     result = parse_reading(s)
    |   File "C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7\weather_alert.py", line 3, in parse_reading
    |     return float(temp), int(humidity)
    |            ~~~~~^^^^^^
    | ValueError: could not convert string to float: ''
    | Failing test case: test_parse_reading_no_crash(
    |     s='/',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7\test_weather_alert.py", line 14, in test_parse_reading_no_crash
    |     result = parse_reading(s)
    |   File "C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7\weather_alert.py", line 2, in parse_reading
    |     temp, humidity = text.split('/')
    |     ^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    | Failing test case: test_parse_reading_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7\test_weather_alert.py", line 14, in test_parse_reading_no_crash
    |     result = parse_reading(s)
    |   File "C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7\weather_alert.py", line 2, in parse_reading
    |     temp, humidity = text.split('/')
    |                      ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_parse_reading_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
________________________________________________________________________________________________ test_parse_reading_bad_humidity _________________________________________________________________________________________________

    @given(st.floats(allow_nan=False, allow_infinity=False),
>          st.integers(max_value=-1) | st.integers(min_value=101))
               ^^^

test_weather_alert.py:27:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

temp = 0.0, humidity = -1

    @given(st.floats(allow_nan=False, allow_infinity=False),
           st.integers(max_value=-1) | st.integers(min_value=101))
    def test_parse_reading_bad_humidity(temp, humidity):
>       assert parse_reading(f"{temp}/{humidity}") is None
E       AssertionError: assert (0.0, -1) is None
E        +  where (0.0, -1) = parse_reading('0.0/-1')
E       Failing test case: test_parse_reading_bad_humidity(
```



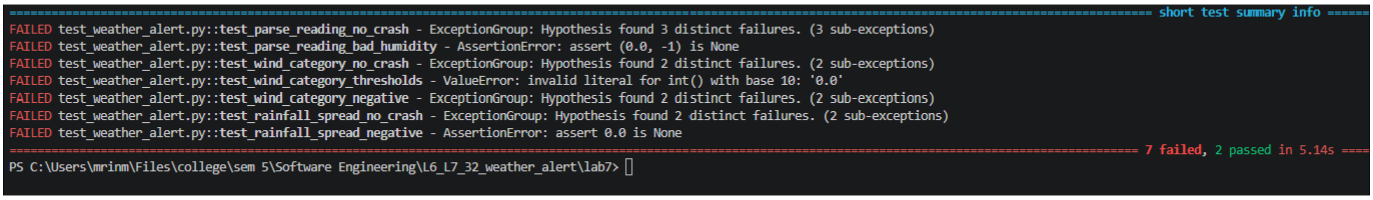
```
E           # The test always failed when commented parts were varied together.
E           temp=0.0,  # or any other generated value
E           humidity=-1,  # or any other generated value
E       )

test_weather_alert.py:29: AssertionError
____________________________________________________________________________ test_wind_category_no_crash ____________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7\test_weather_alert.py", line 36, in test_wind_category_no_crash
  |     def test_wind_category_no_crash(s):
  |                   ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7\test_weather_alert.py", line 37, in test_wind_category_no_crash
    |     assert wind_category(s) in WIND_CATEGORIES | {None}
    |            ~~~~~~~~~~~~~^^^
    |   File "C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7\weather_alert.py", line 6, in wind_category
    |     speed = int(speed_text)
    | ValueError: invalid literal for int() with base 10: ''
    | Failing test case: test_wind_category_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7\test_weather_alert.py", line 37, in test_wind_category_no_crash
    |     assert wind_category(s) in WIND_CATEGORIES | {None}
    |            ~~~~~~~~~~~~~^^^
    |   File "C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7\weather_alert.py", line 6, in wind_category
    |     speed = int(speed_text)
    | TypeError: int() argument must be a string, a bytes-like object or a real number, not 'NoneType'
    | Failing test case: test_wind_category_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
```



```
_____________________________________________________________________________ test_wind_category_thresholds ____

    @given(st.integers(min_value=0, max_value=500) | st.floats(min_value=0, max_value=500))
>   def test_wind_category_thresholds(speed):
                  ^^^

test_weather_alert.py:41:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
test_weather_alert.py:48: in test_wind_category_thresholds
    assert wind_category(str(speed)) == expected
           ^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

speed_text = '0.0'

    def wind_category(speed_text):
>       speed = int(speed_text)
                ^^^^^^^^^^^^^^^
E       ValueError: invalid literal for int() with base 10: '0.0'
E       Failing test case: test_wind_category_thresholds(
E           speed=0.0,
E       )

weather_alert.py:6: ValueError
_____________________________________________________________________________ test_wind_category_negative ____
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7\test_weather_alert.py", line 52, in test_wind_category_negative
  |     def test_wind_category_negative(speed):
  |                   ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7\test_weather_alert.py", line 53, in test_wind_category_negative
    |     assert wind_category(str(speed)) is None
    |            ~~~~~~~~~~~~~^^^^^^^^^^^^
    |   File "C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7\weather_alert.py", line 6, in wind_category
    |     speed = int(speed_text)
    | ValueError: invalid literal for int() with base 10: '-1.0'
    | Failing test case: test_wind_category_negative(
    |     speed=-1.0,  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7\test_weather_alert.py", line 53, in test_wind_category_negative
    |     assert wind_category(str(speed)) is None
    | AssertionError: assert 'none' is None
    |  +  where 'none' = wind_category('-1')
    |  +    where '-1' = str(-1)
    | Failing test case: test_wind_category_negative(
    |     speed=-1,  # or any other generated value
    | )
    +------------------------------------
```

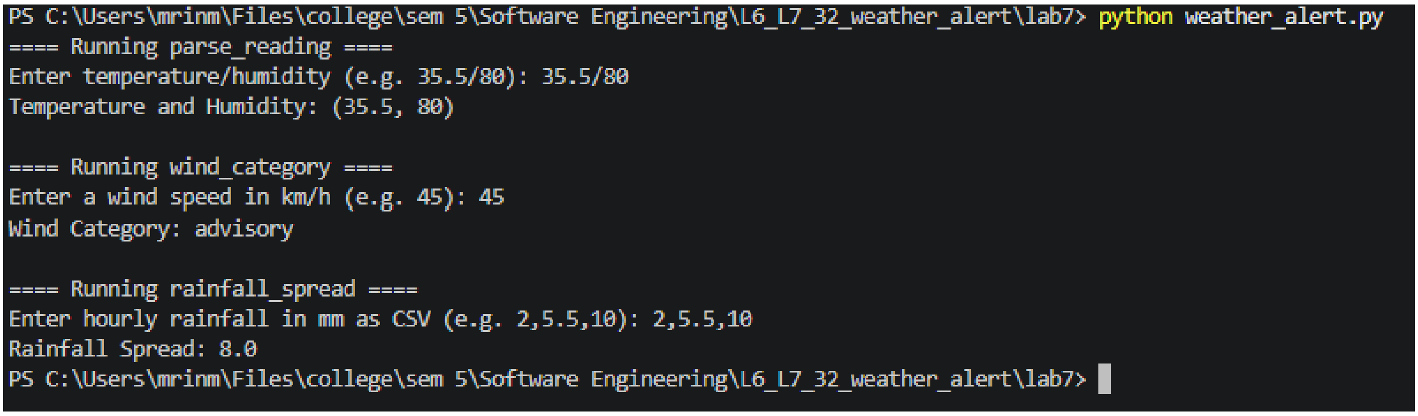


```
___________________________________________________________________________________________________ test_rainfall_spread_no_crash ___________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7\test_weather_alert.py", line 60, in test_rainfall_spread_no_crash
  |     def test_rainfall_spread_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7\test_weather_alert.py", line 61, in test_rainfall_spread_no_crash
    |     result = rainfall_spread(s)
    |   File "C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7\weather_alert.py", line 14, in rainfall_spread
    |     readings = [float(r) for r in readings_csv.split(',')]
    |                 ~~~~~^^^
    | ValueError: could not convert string to float: ''
    | Failing test case: test_rainfall_spread_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7\test_weather_alert.py", line 61, in test_rainfall_spread_no_crash
    |     result = rainfall_spread(s)
    |   File "C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7\weather_alert.py", line 14, in rainfall_spread
    |     readings = [float(r) for r in readings_csv.split(',')]
    |                                   ^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_rainfall_spread_no_crash(
    |     s=None,
    | )
    +------------------------------------
___________________________________________________________________________________________________ test_rainfall_spread_negative ___________________

    @given(st.lists(st.floats(min_value=0, max_value=1e6)),
>          st.floats(max_value=-0.001, allow_infinity=False))
                ^^^

test_weather_alert.py:74:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

readings = [], negative = -1.0

    @given(st.lists(st.floats(min_value=0, max_value=1e6)),
           st.floats(max_value=-0.001, allow_infinity=False))
    def test_rainfall_spread_negative(readings, negative):
        csv = ','.join(str(r) for r in readings + [negative])
>       assert rainfall_spread(csv) is None
E       AssertionError: assert 0.0 is None
E        +  where 0.0 = rainfall_spread('-1.0')
E       Failing test case: test_rainfall_spread_negative(
E           # The test always failed when commented parts were varied together.
E           readings=[],  # or any other generated value
E           negative=-1.0,  # or any other generated value
E       )

test_weather_alert.py:77: AssertionError
```

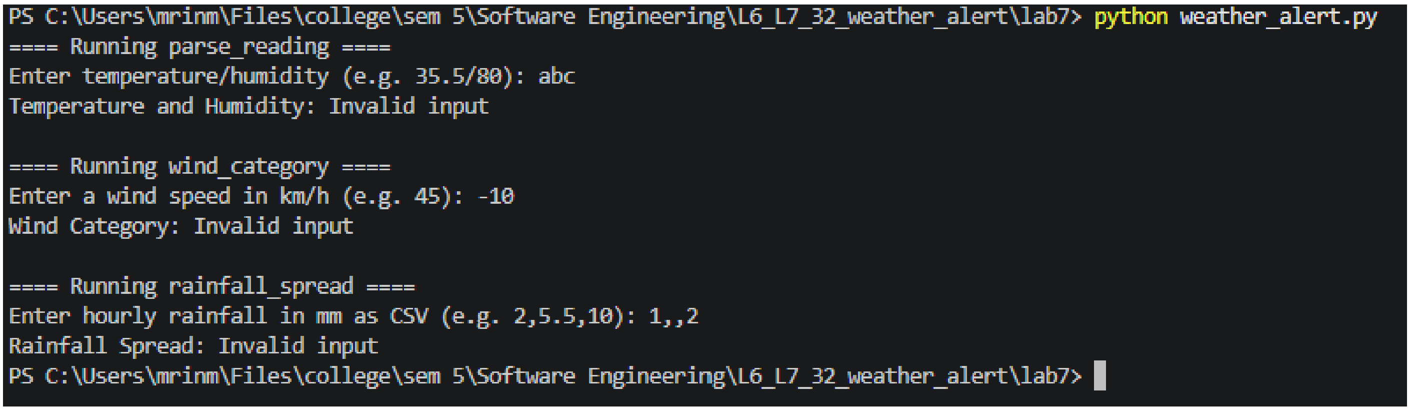


```
=========================================================================================================================== short test summary info ======
FAILED test_weather_alert.py::test_parse_reading_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_weather_alert.py::test_parse_reading_bad_humidity - AssertionError: assert (0.0, -1) is None
FAILED test_weather_alert.py::test_wind_category_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_weather_alert.py::test_wind_category_thresholds - ValueError: invalid literal for int() with base 10: '0.0'
FAILED test_weather_alert.py::test_wind_category_negative - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_weather_alert.py::test_rainfall_spread_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_weather_alert.py::test_rainfall_spread_negative - AssertionError: assert 0.0 is None
========================================================================================================================== 7 failed, 2 passed in 5.14s ====
PS C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7> 
```

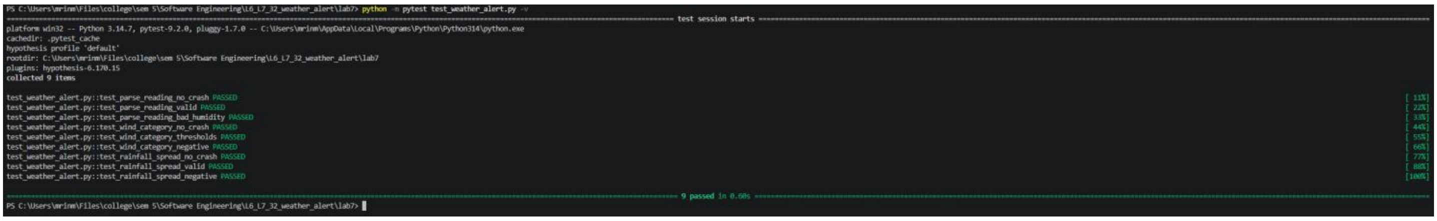
Fixed application:



```
PS C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7> python weather_alert.py
==== Running parse_reading ====
Enter temperature/humidity (e.g. 35.5/80): 35.5/80
Temperature and Humidity: (35.5, 80)

==== Running wind_category ====
Enter a wind speed in km/h (e.g. 45): 45
Wind Category: advisory

==== Running rainfall_spread ====
Enter hourly rainfall in mm as CSV (e.g. 2,5.5,10): 2,5.5,10
Rainfall Spread: 8.0
PS C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7> 
```



```
PS C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7> python weather_alert.py
==== Running parse_reading ====
Enter temperature/humidity (e.g. 35.5/80): abc
Temperature and Humidity: Invalid input

==== Running wind_category ====
Enter a wind speed in km/h (e.g. 45): -10
Wind Category: Invalid input

==== Running rainfall_spread ====
Enter hourly rainfall in mm as CSV (e.g. 2,5.5,10): 1,,2
Rainfall Spread: Invalid input
PS C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7> 
```



```
PS C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7> python -m pytest test_weather_alert.py -v
=========================================================================== test session starts ===========================================================================
platform win32 -- Python 3.14.7, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\mrinm\AppData\Local\Programs\Python\Python314\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7
plugins: hypothesis-6.170.15
collected 9 items

test_weather_alert.py::test_parse_reading_no_crash PASSED                                                                                                          [ 11%]
test_weather_alert.py::test_parse_reading_valid PASSED                                                                                                             [ 22%]
test_weather_alert.py::test_parse_reading_bad_humidity PASSED                                                                                                      [ 33%]
test_weather_alert.py::test_wind_category_no_crash PASSED                                                                                                          [ 44%]
test_weather_alert.py::test_wind_category_thresholds PASSED                                                                                                        [ 55%]
test_weather_alert.py::test_wind_category_negative PASSED                                                                                                          [ 66%]
test_weather_alert.py::test_rainfall_spread_no_crash PASSED                                                                                                        [ 77%]
test_weather_alert.py::test_rainfall_spread_valid PASSED                                                                                                           [ 88%]
test_weather_alert.py::test_rainfall_spread_negative PASSED                                                                                                        [100%]

============================================================================ 9 passed in 0.60s ============================================================================
PS C:\Users\mrinm\Files\college\sem 5\Software Engineering\L6_L7_32_weather_alert\lab7> 
```